PHARMACOLOGY · MCQ PRACTICE

Treatment of GI Parasitic Infestations — MCQ Practice

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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Metronidazole kills which form of Entamoeba histolytica? EASY

A. Cysts only
B. Trophozoites only
C. Both cysts and trophozoites equally
D. Neither — it is inactive against amebae
E. Only the extraintestinal cyst form

2. Which drug is the drug of choice for asymptomatic luminal (cyst-passer) E. histolytica infection? EASY

A. Metronidazole
B. Chloroquine
C. Diloxanide furoate
D. Praziquantel
E. Ivermectin

3. Which drug is the drug of choice for onchocerciasis (river blindness)? EASY

A. Praziquantel
B. Ivermectin
C. Niclosamide
D. Albendazole
E. Pyrantel pamoate

4. Benzimidazoles (albendazole, mebendazole) work by which mechanism? EASY

A. Opening glutamate-gated chloride channels
B. Binding parasite β-tubulin and blocking microtubule polymerisation
C. Increasing membrane calcium permeability
D. Depolarising neuromuscular blockade
E. Inhibiting anaerobic energy metabolism

5. A patient with an amebic liver abscess is started on metronidazole. Why is a second luminal-only agent (e.g. diloxanide furoate or paromomycin) typically added after treating the acute abscess? MEDIUM

A. To treat drug resistance in the liver
B. Metronidazole clears tissue trophozoites but doesn't eliminate luminal cysts, so a luminal agent is needed for a definitive cure and to prevent ongoing transmission
C. Metronidazole has no effect on amebic liver abscess at all
D. The second drug is only for pain control
E. Luminal agents treat the hepatic disease better than metronidazole

6. A pregnant patient is diagnosed with giardiasis. Why might paromomycin be favoured over metronidazole in this scenario? MEDIUM

A. Paromomycin is more effective against giardia

B. Paromomycin is not absorbed from the gut and is considered a safe choice in pregnancy, avoiding metronidazole's theoretical concerns

C. Metronidazole cannot treat giardiasis at all

D. Paromomycin has a faster onset of action

E. Metronidazole is contraindicated in all adult women

7. A patient taking metronidazole for trichomoniasis drinks alcohol at a party and develops flushing, nausea, and tachycardia. What is the mechanism of this reaction? MEDIUM

A. Direct hepatotoxicity from alcohol and metronidazole combined

B. A disulfiram-like reaction from metronidazole's inhibition of aldehyde dehydrogenase-related alcohol metabolism

C. An IgE-mediated allergic reaction to alcohol

D. Serotonin syndrome from metronidazole and alcohol

E. This is unrelated to metronidazole and purely alcohol intoxication

8. A patient on warfarin is started on metronidazole for an anaerobic infection. What should be monitored closely? MEDIUM

A. Increased warfarin effect (metronidazole potentiates warfarin, requiring INR monitoring)

B. Decreased warfarin effect requiring a dose increase

C. No interaction exists between these drugs

D. Warfarin should be stopped entirely and cannot be resumed

E. Only renal function needs monitoring

9. A patient taking phenobarbitone for seizures is started on metronidazole for a dental anaerobic infection. What effect would phenobarbitone (an enzyme inducer) have on metronidazole's efficacy? MEDIUM

A. No effect, since metronidazole is renally cleared

B. Phenobarbitone induces metronidazole metabolism, reducing its effect

C. Phenobarbitone inhibits metronidazole metabolism, increasing toxicity

D. Phenobarbitone has no interaction with any GI pharmacology drug

E. This combination causes a disulfiram-like reaction unrelated to alcohol

10. A patient with ocular cysticercosis is mistakenly prescribed praziquantel without ophthalmologic evaluation. What is the specific danger of this approach? MEDIUM

A. Praziquantel has no effect on cysticercosis

B. Killing the organism in the eye with praziquantel can cause irreversible ocular damage from the resulting inflammatory response

C. Praziquantel would cause systemic anaphylaxis unrelated to the eye

D. This combination causes fatal cardiac arrhythmia

E. Praziquantel would cure the eye infection with no risk

11. A patient is treated with ivermectin for strongyloidiasis. Which pharmacokinetic property explains why ivermectin has minimal CNS side effects despite its neuromuscular mechanism in the parasite? MEDIUM

A. Ivermectin is completely inactive in mammals

B. Ivermectin does not readily cross the mammalian blood-brain barrier, unlike in nematodes where it opens glutamate-gated chloride channels in nerve/muscle

C. Ivermectin only works on parasites without a nervous system

D. Ivermectin is entirely metabolized before reaching the brain regardless of BBB permeability

E. Ivermectin has no mechanism involving chloride channels

12. A patient with cryptosporidiosis and AIDS is more likely to require hospitalization than an immunocompetent patient with the same infection. What does this reveal about cryptosporidiosis in general? MEDIUM

A. Cryptosporidiosis is always fatal regardless of immune status

B. Cryptosporidiosis is typically self-limited in immunocompetent hosts but can be severe in immunocompromised patients

C. Cryptosporidiosis only occurs in immunocompromised patients

D. Nitazoxanide is contraindicated in immunocompetent patients

E. Immune status has no bearing on cryptosporidiosis severity

13. A patient is treated with praziquantel for schistosomiasis and is also taking phenytoin for seizures. What effect would phenytoin have on praziquantel levels? MEDIUM

A. Phenytoin has no interaction with praziquantel

B. Phenytoin (a CYP inducer) would decrease praziquantel levels, reducing efficacy

C. Phenytoin would increase praziquantel levels, causing toxicity

D. Praziquantel would inactivate phenytoin

E. This combination is contraindicated due to fatal interaction

14. Considering that diloxanide furoate, iodoquinol, and paromomycin are all luminal-only amebicides, why would none of them be appropriate as monotherapy for a patient with active amebic colitis (invasive tissue disease)? MEDIUM

A. They are all contraindicated in any GI infection

B. They act only within the intestinal lumen and have no activity against invasive tissue trophozoites, so they cannot address the tissue-invasive component of colitis

C. They are only approved for use in pregnancy

D. They cause the same disulfiram-like reaction as metronidazole

E. They are more toxic than metronidazole in tissue disease

15. A patient has confirmed E. histolytica colitis with bloody diarrhea. The clinician wants to eliminate both the invasive tissue trophozoites and the luminal cyst-passing state to prevent relapse and transmission. Based on the mechanisms described, which combination best achieves both goals? HARD

A. Diloxanide furoate alone

B. Metronidazole (tissue-active, kills trophozoites) followed by/combined with a luminal agent such as iodoquinol or diloxanide furoate (clears luminal cysts), since neither drug type alone covers both compartments

C. Chloroquine alone, since it treats both tissue and luminal disease

D. Ivermectin, since it has broad amebicidal activity

E. Niclosamide, since it treats both cestode and protozoal infection

16. A researcher notes that ivermectin is highly effective in ascariasis/enterobiasis but comparatively less effective in trichuriasis (whipworm) and hookworm. Meanwhile, benzimidazoles are first-line specifically for whipworm, pinworm, hookworm, and roundworm. What does this pattern suggest about drug selection strategy for mixed intestinal helminth infections? HARD

A. Ivermectin and benzimidazoles are fully interchangeable for all nematode infections

B. Drug choice should be tailored to the specific worm species present, since efficacy varies meaningfully across nematode species even within the same broad drug class comparison

C. Only ivermectin should ever be used for any nematode infection

D. Only benzimidazoles should ever be used, since ivermectin has no clinical utility

E. Species-specific efficacy differences described here are clinically irrelevant

17. A patient with hydatid disease (Echinococcus) is being treated with albendazole. A colleague suggests niclosamide might work just as well since "they're both anthelmintics." Why is this substitution inappropriate based on the lecture's content? HARD

A. Niclosamide and albendazole have identical spectra, so the substitution is appropriate

B. Niclosamide is used for taeniasis/diphyllobothriasis as an alternative to praziquantel, not for cysticercosis/hydatid disease, which is albendazole's specific niche — the two drugs are not interchangeable across all cestode-related indications

C. Niclosamide is more effective for hydatid disease than albendazole

D. Both drugs are contraindicated in hydatid disease

E. Niclosamide is a protozoal drug, not an anthelmintic, so this comparison doesn't apply

18. A patient with schistosomiasis and a history of ocular cysticercosis (previously treated and resolved) needs treatment for their current schistosome infection. Praziquantel is being considered. What is the key clinical reasoning point here? HARD

A. Since the cysticercosis is already resolved, praziquantel can be used for the current schistosomiasis without any special precaution related to the eye

B. Praziquantel remains contraindicated in any patient with active ocular cysticercosis specifically — the eye-related contraindication applies to active ocular disease being treated, not to a fully resolved prior episode, so current ocular status (not history alone) should be confirmed before treatment

C. Praziquantel can never be used in any patient with any history of cysticercosis, active or resolved, under any circumstances

D. Praziquantel is only contraindicated for schistosomiasis, not cysticercosis

E. This scenario has no relevance to praziquantel's contraindications

19. A patient is diagnosed with both giardiasis and a co-existing anaerobic dental abscess. The clinician selects metronidazole as a single agent for both conditions. What pharmacological principle explains why this dual-purpose choice is reasonable? HARD

A. Metronidazole has a spectrum that includes both protozoa (Giardia, E. histolytica, Trichomonas) and anaerobic bacteria (via the same prodrug-activation mechanism), making it appropriate for concurrent protozoal and anaerobic bacterial infections

B. Metronidazole only treats bacterial infections, and the giardiasis improvement would be coincidental

C. Metronidazole cannot treat both conditions simultaneously and would need to be given as two separate courses

D. This combination is dangerous and contraindicated

E. Giardia and anaerobic bacteria require entirely different drug classes with no overlap

20. A student notes that nitazoxanide is used for cryptosporidiosis, giardiasis, amebiasis, AND some intestinal helminths — a broader spectrum than most other agents discussed. What is the most likely explanation for this breadth, based on its described mechanism? HARD

A. Nitazoxanide works by a completely random, non-specific toxic mechanism unrelated to any conserved target

B. Nitazoxanide inhibits an anaerobic energy metabolism enzyme, a metabolic pathway that is conserved across diverse anaerobic/microaerophilic protozoa and some helminths, explaining its broader spectrum compared to agents targeting more organism-specific structures

C. Nitazoxanide's spectrum is a labeling error and it is not actually broad-spectrum

D. Nitazoxanide works only on protozoa and has no helminth activity, contradicting the premise

E. Nitazoxanide's spectrum comes from simultaneous action on human immune cells, not the parasites themselves

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	Metronidazole kills trophozoites only, not cysts.
2	C	EASY	Diloxanide furoate is the drug of choice for asymptomatic luminal infection.
3	B	EASY	Ivermectin is the drug of choice for onchocerciasis, killing microfilariae (not adult worms).
4	B	EASY	Benzimidazoles bind β-tubulin, blocking microtubule polymerisation and glucose uptake.
5	B	MEDIUM	Metronidazole is effective for tissue disease but does not reliably clear luminal cysts; a luminal amebicide is added for a definitive cure.
6	B	MEDIUM	Paromomycin is not absorbed from the gut and is noted as a safe choice in pregnancy, unlike the general caution around metronidazole/tinidazole.
7	B	MEDIUM	Metronidazole and tinidazole cause a disulfiram-like reaction with alcohol (nausea, flushing, tachycardia, hypotension).

8	**A**	MEDIUM	Metronidazole increases warfarin's effect, so INR should be monitored closely when the two are combined.
9	**B**	MEDIUM	Enzyme inducers like phenobarbitone/rifampin decrease metronidazole's effect via increased hepatic metabolism.
10	**B**	MEDIUM	Praziquantel is contraindicated in ocular cysticercosis because killing the organism in the eye can cause irreversible damage from the resulting inflammatory response.
11	**B**	MEDIUM	Ivermectin opens glutamate-gated chloride channels in nematode nerve/muscle causing paralysis, but does not readily cross the mammalian blood-brain barrier, limiting CNS side effects in the host.
12	**B**	MEDIUM	Cryptosporidiosis is self-limited if immunocompetent but can require hospitalization in AIDS/immunocompromised patients.
13	**B**	MEDIUM	CYP inducers like phenytoin, carbamazepine, and phenobarbital decrease praziquantel levels.
14	**B**	MEDIUM	Luminal-only amebicides have no activity against tissue-invasive trophozoites, so they cannot treat the invasive component of colitis alone — a tissue-active agent like metronidazole is needed first.
15	**B**	HARD	Since metronidazole kills trophozoites (tissue-active) but not cysts, and luminal agents like iodoquinol/diloxanide act only in the lumen, a combination addressing both compartments is needed for complete cure and to prevent luminal cyst-passing/relapse.
16	**B**	HARD	The differential efficacy (ivermectin better for ascariasis/enterobiasis and strongyloidiasis/onchocerciasis; benzimidazoles first-line for whipworm/pinworm/hookworm/roundworm) means empiric drug selection should consider which specific worm(s) are involved rather than using either agent universally.
17	**B**	HARD	While both are used for tapeworm-related conditions, albendazole's described primary use is cysticercosis and hydatid disease, while niclosamide is described as an alternative to praziquantel specifically for taeniasis and diphyllobothriasis — they are not interchangeable across all indications.
18	**B**	HARD	The contraindication specifically concerns active ocular cysticercosis (killing the organism in the eye causing irreversible damage) — the clinical reasoning point is to confirm current ocular status rather than assume a resolved historical episode changes the risk calculus if any residual ocular organisms were present.
19	**A**	HARD	Metronidazole's spectrum explicitly includes both protozoa (E. histolytica, Giardia lamblia, Trichomonas vaginalis) and anaerobic bacteria, all via the same nitroreductase-activated prodrug mechanism, making single-agent treatment of concurrent infections from both categories pharmacologically sound.
20	**B**	HARD	Nitazoxanide inhibits an anaerobic energy metabolism enzyme — a metabolic dependency shared across diverse anaerobic/microaerophilic protozoa and certain helminths — which plausibly explains its comparatively broader described spectrum relative to more organism-specific agents (e.g. benzimidazoles targeting β-tubulin).